Short-Burst Optimization for Minimum-Edge-Cut Redistricting

G.6 — Ensemble Methods / Algorithm Comparison

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Abstract

Short-Burst optimization [Cannon et al., 2022] applies sequences of short ReCom chains—each restarting from the *endpoint* of the previous burst—to concentrate exploration near low-edge-cut plans while preserving the Markov property. We implement Short-Burst in `redist-cli` as `SeedCompositor::ShortBurst` and compare it to multi-seed METIS, ConvergenceSweep, and BisectionEnsemble on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) under a fixed budget of 1,000 total ReCom steps. ShortBurst achieves Polsby-Popper improvements of 15–22% over the standard single-seed METIS baseline, placing it between multi-seed METIS (8–12%) and BisectionEnsemble (15–26%). The key correctness requirement is that each burst retains its *endpoint*—not the within-burst minimum—to preserve the Markov restart property and avoid systematic bias toward lower-edge-cut plans in a way incompatible with the stationary distribution. Sensitivity analysis on NC shows that burst length governs performance more strongly than burst count: bursts of 5–20 steps outperform bursts of 50+ steps, which drift and resemble single-chain ReCom.

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1. Introduction

1.1. The Problem: METIS Finds Good but Not Optimal Plans

METIS multilevel graph partitioning [Della-Libera, 2026a] produces redistricting plans with low edge-cut in $O(n \log n)$ time. For a state with $n \approx 2{,}000$ census tracts and k districts, a single METIS run typically achieves Polsby-Popper compactness at the 70th–80th percentile of the feasible plan distribution. But METIS is a local optimizer: it terminates at a local minimum of the edge-cut objective, and different random seeds reach different local minima.

Multi-seed METIS (the `-search multi` mode) runs m independent METIS seeds and returns the best plan. With $m = 50$ seeds, multi-seed METIS achieves 8–12% Polsby-Popper improvement over single-seed. But the improvement saturates quickly: seeds are independent and explore different basins rather than the neighborhood of any one good plan.

Long ReCom chains (e.g., the ConvergenceSweep algorithm with $T = 600$ steps) achieve stronger mixing and can escape local minima [Della-Libera, 2026b]. However, a single long chain explores broadly, spending many steps in regions of plan space far from the best-found plan. The chain cannot "remember" and return to the best plan it visited.

1.2. Short-Burst: Concentrated Neighborhood Exploration

Short-Burst optimization [Cannon et al., 2022] addresses this problem directly. The key idea is to decompose the total step budget into a sequence of *short bursts*—each burst is a short ReCom chain that starts from the *endpoint* of the previous burst. After each burst, the endpoint is recorded and becomes the starting point for the next burst. The final plan is selected from the recorded endpoints by a percentile criterion.

The restart mechanism means the chain never drifts far from the best-found plan: each burst explores a small neighborhood, and the best neighborhood endpoint becomes the new center for the next exploration. This concentrates probability near low-edge-cut regions of plan space without sacrificing the Markov property.

Cannon et al. [Cannon et al., 2022] introduced Short-Burst in the context of partisan outcome analysis—using short chains to measure partisan sensitivity of redistricting plans. Our use of Short-Burst for minimum-edge-cut compactness optimisation is a natural extension: we apply the same restart mechanism to the Polsby-Popper objective rather than a partisan metric.

1.3. Main Contributions

This paper makes three contributions:

1. **Implementation.** We implement ShortBurst in `redist-cli`. The search mode is `-search short-burst`; the Rust type is `SeedCompositor::ShortBurst`. Per-burst seeds are derived by SHA-256 from the base seed and burst index, ensuring reproducibility across runs.

2. **Empirical comparison.** We compare ShortBurst to multi-seed METIS, ConvergenceSweep, and BisectionEnsemble on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) under a fixed budget of 1,000 total ReCom steps. ShortBurst achieves 15–22% Polsby-Popper improvement, placing it between multi-seed METIS and BisectionEnsemble.

3. **Sensitivity analysis.** We characterise how burst length ℓ and burst count m interact. Burst length is the critical parameter: short bursts ($\ell = 5$–20) outperform long bursts ($\ell = 50+$), which drift and approach single-chain ReCom in behaviour.

1.4. Paper Structure

Section 2 reviews ReCom, ConvergenceSweep, BisectionEnsemble, and the original Short-Burst paper. Section 3 gives the formal algorithm and the chain-seed derivation. Section 4 presents the four-way comparison on NC, WI, and TX. Section 5 gives the burst-length and burst-count sensitivity analysis. Section 6 concludes.

2. Background

2.1. ReCom

The Recombination (ReCom) Markov chain [DeFord et al., 2021] is the standard Markov chain Monte Carlo method for redistricting. Given a current plan π on census-tract graph $G = (V, E)$:

1. Select a random boundary edge (u, v) with $\pi(u) \neq \pi(v)$.

2. Merge districts $A = \pi^{-1}(\pi(u))$ and $B = \pi^{-1}(\pi(v))$ into the subgraph $G[A \cup B]$.
3. Sample a random spanning tree $\mathcal{T}$ of $G[A \cup B]$.
4. Find an edge in $\mathcal{T}$ whose removal partitions $A \cup B$ into two population-balanced subsets; if none exists, reject.
5. Update π with the new bipartition.

ReCom is designed to have a stationary distribution approximately uniform over valid plans (connected, population-balanced bipartitions). It is the foundational chain for all methods discussed in this paper.

2.2. ConvergenceSweep

ConvergenceSweep [Della-Libera, 2026b] runs a single ReCom chain for T steps and returns the best plan seen during the run. The parameter $T = 600$ is calibrated as the statutory stopping threshold: chains run beyond $T = 600$ steps show diminishing Polsby-Popper improvement (less than 1% per additional 100 steps) on all 50 states.

ConvergenceSweep operates with a fixed step budget and selects the minimum-edge-cut plan encountered during the entire run. Selecting the within-run minimum is appropriate for ConvergenceSweep because the chain is run as a single continuous trajectory—there is no restart mechanism, and the chain is not being used as a Markov restart point.

2.3. BisectionEnsemble

BisectionEnsemble [Della-Libera, 2026d] applies ReCom ensembles at each node of the ApportionRegions bisection tree. Rather than running a single ensemble on the full state, BisectionEnsemble runs a short ensemble ($T = 100$ steps) at each bisection node, selecting the best endpoint at each node. The node-level best plan propagates down the tree.

BisectionEnsemble with $p = 0.0$ (always select the best endpoint) achieves 15–26% Polsby-Popper improvement over single-seed METIS across NC, WI, and TX, and is the strongest benchmark in our comparison.

2.4. Short-Burst: Original Paper

Cannon, Duchin, Randall, and Rule [Cannon et al., 2022] introduced Short-Burst optimization in the context of partisan outcome analysis. Their setting: given an enacted redistricting plan, measure its partisan properties by running short ReCom chains from the enacted plan and examining the distribution of partisan outcomes at chain endpoints.

Short bursts of length ℓ starting from a given plan sample the *local neighbourhood* of that plan in plan space. By restarting from each endpoint rather than the initial plan, successive bursts explore the neighbourhood of the neighbourhood, gradually drifting toward better plans without the slow mixing of a long chain.

Endpoint vs. minimum. A critical design choice distinguishes Short-Burst from naive greedy chain-and-pick: Cannon et al. retain the *endpoint* of each burst, not the within-burst minimum.

This distinction has two consequences:

1. **Markov property.** The endpoint of a ReCom chain is a valid plan in $\mathcal{P}$ that could be the starting point of a new chain with the correct Markov structure. The within-burst minimum is not a valid restart point in general: it was not the chain's terminal state, and the chain did not end in that state—restarting from it would break the Markov property and introduce path dependencies not accounted for in the stationary distribution.

2. **Stationarity.** Selecting the minimum within each burst would systematically bias the restarted chain toward lower-edge-cut plans in a way not captured by the chain's stationary distribution. The bias is a form of selection pressure: the chain preferentially transitions to lower-cut plans not because the transition probability is higher, but because the selection mechanism discards higher-cut endpoints. This makes the selected sequence a biased chain, not a valid Markov chain on $\mathcal{P}$.

In our min-EC compactness setting, the same argument applies: we retain burst endpoints and select among them using the percentile parameter p [Della-Libera, 2026c].

From partisan analysis to compactness optimisation. Cannon et al. used Short-Burst to measure partisan sensitivity: how much does a plan's partisan lean vary in its local neighbourhood? Our use is different: we use Short-Burst as an optimization algorithm to find low-edge-cut plans. The mechanism is identical; only the objective differs. This is a natural extension—the endpoint-retention and restart properties are valuable regardless of the objective being optimized.

3. Algorithm

3.1. Formal Definition

Let $G = (V, E)$ be a census-tract adjacency graph with $|V| = n$, k the number of districts, and $\mathrm{EC}(\pi)$ the edge-cut of plan π (the number of edges in E crossing district boundaries). Equivalently, $1 - \mathrm{EC}(\pi)/|E|$ is monotone in Polsby-Popper: lower edge-cut corresponds to higher compactness.

Definition 1 (ShortBurst)**.** *Given parameters ℓ (burst length), m (burst count), $p \in [0, 1]$ (selection percentile), and a base seed $s \in \{0, \dots, 2^{64} - 1\}$:*

1. *Compute the initial plan $\pi_0 = \mathrm{METIS}(G, k, s)$.*
2. *For $i = 0, \dots, m - 1$:*
   (a) *Derive the burst seed $s_i = \mathrm{chain\_seed}(s, i)$.*
   (b) *Initialize a ReCom chain from π_i using s_i as the RNG seed.*
   (c) *Run the chain for ℓ steps to obtain $\pi_i^{(\ell)}$.*
   (d) *Record the endpoint: $e_i = (\mathrm{EC}(\pi_i^{(\ell)}),\ i,\ \pi_i^{(\ell)})$.*
   (e) *Set $\pi_{i+1} = \pi_i^{(\ell)}$ (restart from endpoint).*
3. *Sort endpoints: $e_{(0)} \leq e_{(1)} \leq \cdots \leq e_{(m-1)}$ by (EC, i) lexicographically (ascending EC, then burst index as tiebreak).*
4. *Return $e_{(\lfloor p \cdot m \rfloor)}.\pi$ (the plan at the p-th quantile of recorded endpoints by edge-cut).*

The default parameters in `redist-cli` are $\ell = 20$, $m = 50$, $p = 0.0$ (always return the best endpoint found).

3.2. Chain-Seed Derivation

Each burst uses a deterministic, independent RNG seed derived from the base seed and burst index. This ensures:

- **Reproducibility**: given the same base seed, the same sequence of plans is produced on any run.
- **Independence**: burst seeds are cryptographically independent, preventing correlation between consecutive burst RNG streams.
- **No seed reuse**: each burst index maps to a unique seed, preventing the same random choices from appearing in different bursts.

The derivation uses SHA-256. Define the domain-separated byte string:

$$D(s,i) \;=\; \underbrace{\texttt{"SB\_CHAIN\_"}}_{\text{9 bytes}} \;\|\; \underbrace{i.\text{to\_le\_bytes}()}_{\text{8 bytes}} \;\|\; \underbrace{\texttt{"\_"}}_{\text{1 byte}} \;\|\; \underbrace{s.\text{to\_le\_bytes}()}_{\text{8 bytes}},$$

then:

$$\text{chain\_seed}(s,i) = \text{SHA256}(D(s,i))[0..8],$$

where [0..8] denotes the first 8 bytes of the SHA-256 digest, interpreted as a little-endian `u64`. The prefix serves as domain separator; the implementation uses the longer string `"SHORT_BURST_CHAIN_"` for legibility.

This is the same domain-separation pattern used in BisectionEnsemble [Della-Libera, 2026d] and ConvergenceSweep [Della-Libera, 2026b]: each algorithm uses a unique ASCII prefix to ensure no seed collision across search modes.

3.3. Pseudocode

Algorithm 1 ShortBurst Plan Selection

Require: Graph G, districts k, burst length ℓ, burst count m, percentile p, base seed s

Ensure: A redistricting plan π^*

$\pi_0 \leftarrow \text{METIS}(G, k, s)$
endpoints $\leftarrow$ []
for $i \leftarrow 0$ **to** $m - 1$ **do**
    $s_i \leftarrow \text{chain\_seed}(s, i)$
    rng $\leftarrow$ SmallRng :: seed_from_u64(s_i)
    chain $\leftarrow$ RecomChain($\pi_i,\ G,\ k$)
    for $j \leftarrow 0$ **to** $\ell - 1$ **do**
        chain.step(*rng*)
    end for
    $\pi_{i+1} \leftarrow$ *chain*.assignment ▷ endpoint, not minimum
    endpoints.push((EC(π_{i+1}), i, π_{i+1}))
end for
endpoints.sort_by(EC asc, i asc) ▷ stable, deterministic
return *endpoints*[$\lfloor p \cdot m \rfloor$].$\pi$

3.4. CLI Invocation

```
redist state --state NC --year 2020
    --search short-burst
    --burst-length 20
    --n-bursts 50
    --percentile 0.0
```

The percentile parameter `-percentile 0.0` selects the best endpoint (minimum edge-cut) across all bursts. Setting `-percentile 0.5` returns the median endpoint, which may be preferable when statistical properties of the neighbourhood are of interest rather than the optimum.

3.5. Stochastic Dominance

Theorem 1 (Short-Burst Stochastic Dominance)**.** *Let $\pi^{\textsc{ShortBurst}}$ be the plan returned by* ShortBurst *with ℓ steps per burst and m bursts, and let $\pi^{\textsc{ReCom}}$ be the plan at the end of a single* ReCom *chain of $\ell \cdot m$ steps (same total step budget), both starting from the same METIS initialisation. Then for any threshold $c > 0$:*

$$\Pr\Big[\mathrm{EC}(\pi^{\textsc{ShortBurst}}) \leq c\Big] \;\geq\; \Pr\Big[\mathrm{EC}(\pi^{\textsc{ReCom}}) \leq c\Big].$$

That is, ShortBurst *stochastically dominates single-chain* ReCom *of the same total length for finding low-edge-cut plans.*

Proof sketch. The key observation is that after burst i, the chain restarts from π_{i+1}, which has already been selected as the best-available starting point for the next burst. By contrast, the single chain at step $i \cdot \ell$ has no such restart—it may be at a higher-edge-cut state than any endpoint recorded so far.

Formally, let $\mathcal{E}_i = \min_{j \leq i} \mathrm{EC}(\pi_j^{(\ell)})$ be the best edge-cut seen after i bursts. For the single chain, let $\mathcal{E}'_T = \mathrm{EC}(\pi'_T)$ be the edge-cut at step $T = i \cdot \ell$.

After i bursts, the ShortBurst chain is at state π_{i+1} with $\mathrm{EC}(\pi_{i+1}) \geq \mathcal{E}_i$ (the restart is from the endpoint, not the best). However, the next burst starts from π_{i+1}, which is constrained to have $\mathrm{EC}(\pi_{i+1}) \leq \mathrm{EC}(\pi_i^{(\ell)}) + \Delta$ for some bounded drift Δ over ℓ steps.

By induction on the number of bursts: the restart mechanism ensures that the chain cannot drift far from the best-found plan. The $p = 0.0$ selection then returns the minimum over all burst endpoints, which is at least as good as any single endpoint of the same-budget chain. □

Remark 1. *This is a heuristic dominance claim: the proof sketch above bounds the drift per burst but does not formally characterise the mixing time of the local neighbourhood. A formal proof would require mixing time bounds on the restriction of the* ReCom *chain to the δ-neighbourhood of a good plan—a technically involved argument beyond the scope of this paper. The empirical evidence in Section 4 provides strong evidence for the dominance claim in practice.*

3.6. Why Endpoint Retention is Non-Negotiable

We emphasise the correctness requirement at the heart of Short-Burst.

Breaking the Markov property. If one were to restart from the within-burst minimum rather than the endpoint, the resulting sequence would not be a Markov chain on $\mathcal{P}$. The within-burst minimum depends on the entire trajectory of the burst, not just the current state. A chain that restarts from trajectory-dependent states violates the Markov property: the future evolution depends on past history beyond the current state.

Systematic bias. Selecting the within-burst minimum introduces systematic bias: the selected plan is systematically lower in edge-cut than the chain's stationary distribution would imply. This is not merely a theoretical concern—it means the selected plan cannot be interpreted as a sample from the stationary distribution, and the sequence of selected plans does not approximate any well-defined distribution on $\mathcal{P}$.

Correct behaviour. Endpoint retention preserves both properties. The endpoint is a valid state of the ReCom chain—the chain ended there, and the next chain starts there, maintaining the Markov property. The sequence of endpoints is a valid stochastic process on $\mathcal{P}$, and the selection of the minimum endpoint is a well-defined post-processing step that does not affect the validity of each individual restart.

4. Empirical Comparison

4.1. Experimental Setup

We compare four search methods under a fixed budget of 1,000 total ReCom steps on three states:

- **NC** (North Carolina): $k = 14$ districts, $n = 2{,}195$ tracts.
- **WI** (Wisconsin): $k = 8$ districts, $n = 1{,}406$ tracts.
- **TX** (Texas): $k = 38$ districts, $n = 5{,}265$ tracts.

All methods use the `prime-factor` (ApportionRegions) structure and `county` weights, which are the empirically best-performing structure and weight settings [Della-Libera, 2026a]. The step budget is allocated as follows:

- **Multi-seed METIS**: $m = 50$ independent METIS seeds; no ReCom steps. The "1,000 steps" budget is not applicable—this method uses METIS rather than ReCom, and is included as the primary non-MCMC baseline.
- **ConvergenceSweep**: single ReCom chain, $T = 1{,}000$ steps, returns the minimum-edge-cut plan encountered during the run. (Note: the standard B.16 threshold is $T = 600$; we use $T = 1{,}000$ to match the total step budget.)
- **ShortBurst**: $\ell = 20$ steps per burst, $m = 50$ bursts ($20 \times 50 = 1{,}000$ total steps), $p = 0.0$. Returns the minimum endpoint across all bursts.
- **BisectionEnsemble**: $T = 100$ steps at each node of the ApportionRegions bisection tree. For NC ($k = 14 = 2 \times 7$), the bisection tree has $\lceil \log_2 14 \rceil = 4$ levels and approximately 7–15 active nodes, giving $\approx$ 700–1,500 total steps. We use $T = 100$ with the ApportionRegions tree structure [Della-Libera, 2026d].

4.2. Polsby-Popper Metric

The Polsby-Popper score for a plan π is:

$$\mathrm{PP}(\pi) = \frac{1}{k} \sum_{i=1}^{k} \frac{4\pi \cdot \mathrm{Area}(D_i)}{\mathrm{Perimeter}(D_i)^2},$$

where D_i is district i. Higher PP indicates more compact districts. All improvements are reported as percentage improvement over the standard single-seed METIS baseline.

4.3. Baseline Values

The baseline single-seed METIS Polsby-Popper scores (from B.11 [Della-Libera, 2026a]) and BisectionEnsemble results (from H.1 [Della-Libera, 2026d]) are:

- NC: baseline PP = 0.217; BisectionEnsemble at $p = 0.0$: PP = 0.261 (+20.3%).
- WI: baseline PP = 0.198; BisectionEnsemble at $p = 0.0$: PP = 0.228 (+15.2%).

- TX: baseline PP = 0.184; BisectionEnsemble at $p = 0.0$: PP = 0.231 (+25.5%).

4.4. Results

Table 1 presents the four-way comparison. All results are single-run; Phase 2 will provide multi-run statistics and confidence intervals.

Table 1: Polsby-Popper (PP) and improvement under four methods. Budget: 1,000 ReCom steps. Baseline PP: NC 0.217, WI 0.198, TX 0.184.

Method	NC	%↑	WI	%↑	TX
Multi-seed METIS	0.237	+9.2	0.214	+8.1	0.202
ConvergenceSweep ($T = 1000$)	0.243	+12.0	0.220	+11.1	0.208
ShortBurst ($\ell = 20$, $m = 50$)	0.254	+17.1	0.231	+16.7	0.221
BisectionEnsemble ($T = 100$)	0.261	+20.3	0.228	+15.2	0.231
%↑ TX: +9.8, +13.0, +20.1, +25.5					

All methods produce proportional partisan outcomes (NC 7D/7R, WI 4D/4R, TX 19D/19R), consistent with the ApportionRegions prime-factor structure.

4.5. Partisan Outcomes

Partisan outcomes are computed using 2020 presidential election returns (two-party vote share) mapped to the generated districts. All four methods produce plans in the proportional range: NC 7D/7R, WI 4D/4R, TX 19D/19R. This is consistent with the ApportionRegions prime-factor structure, which minimises edge-cut without partisan input and naturally produces proportional outcomes [Della-Libera, 2026a].

ShortBurst does not change the partisan structure of the plans: the endpoint-retention mechanism explores the low-edge-cut neighborhood of the initial METIS plan, and the partisan outcomes are determined by the geography and district boundaries, not by the selection mechanism.

4.6. Discussion

Three observations stand out:

Short-Burst vs. ConvergenceSweep. ShortBurst outperforms ConvergenceSweep on all three states (+4.7 to +7.1 percentage points). The reason is the restart mechanism:

ConvergenceSweep spends many steps in regions far from the best plan, while Short-Burst returns to a high-quality neighborhood after each burst.

Short-Burst vs. BisectionEnsemble. BisectionEnsemble outperforms ShortBurst on NC and TX, but ShortBurst slightly outperforms BisectionEnsemble on WI (+16.7% vs +15.2%). BisectionEnsemble applies ensembles at each tree node, concentrating steps at the most structurally important decisions (the early bisections that determine state-wide compactness). ShortBurst operates on the full plan, which may be more beneficial for smaller k where the full plan is more tractable.

WI reversal. The WI reversal (ShortBurst beats BisectionEnsemble) is consistent with the small-k advantage hypothesis: for $k = 8$, the bisection tree has only 3 levels and 7 nodes, giving BisectionEnsemble fewer opportunities to apply concentrated optimization than for larger k. ShortBurst's full-plan neighborhood search is relatively more effective.

Combined strategy. The results suggest a natural combination: use BisectionEnsemble for the tree-level optimization, then apply ShortBurst to the final plan for post-processing. This combination is not evaluated in this paper but is a natural direction for Phase 2.

5. Sensitivity Analysis

5.1. Design

We perform a grid search over burst length $\ell \in \{5, 10, 20, 50\}$ and burst count $m \in \{10, 25, 50, 100\}$ on NC ($k = 14$), holding the base seed fixed ($s = 42$) and reporting single-run Polsby-Popper. Total step budgets range from $\ell \cdot m = 50$ to 5,000 steps.

To disentangle the effect of burst length from total step budget, we also run a fixed-budget comparison at exactly 1,000 total steps:

$$(\ell, m) \in \{(5, 200),\ (10, 100),\ (20, 50),\ (50, 20),\ (100, 10)\}.$$

5.2. Burst Length Sensitivity

Table 2 reports PP as a function of burst length, fixing $m = 50$ (total steps $= 50\ell$).

Table 2: Polsby-Popper on NC as a function of burst length ℓ, with fixed $m = 50$ bursts. Baseline (single-seed METIS): PP = 0.217. Total steps = 50ℓ. Single-run results.

Burst length ℓ	Total steps	PP	Improvement	Notes
5	250	0.248	+14.3%	Short: tight neighbourhood
10	500	0.253	+16.6%	Good balance
20	1,000	0.254	+17.1%	Default setting
50	2,500	0.247	+13.8%	Drift sets in
100	5,000	0.241	+11.1%	Approaches single-chain ReCom

The optimal burst length on NC is $\ell = 20$, with a broad plateau from $\ell = 10$ to $\ell = 20$. Performance degrades at $\ell = 50$ and $\ell = 100$: long bursts allow the chain to drift far from the starting plan, and the endpoint is no longer in a high-quality neighborhood. At $\ell = 100$, the algorithm approaches single-chain ReCom: 50 bursts of 100 steps each is effectively a 5,000-step chain with 50 recorded checkpoints.

5.3. Burst Count Sensitivity

Table 3 reports PP as a function of burst count m, fixing $\ell = 20$ (total steps = $20m$).

Table 3: Polsby-Popper on NC as a function of burst count m, with fixed burst length $\ell = 20$. Baseline: PP = 0.217. Total steps = $20m$. Single-run results.

Burst count m	Total steps	PP	Improvement	Notes
10	200	0.244	+12.4%	Few bursts; limited exploration
25	500	0.250	+15.3%	Moderate
50	1,000	0.254	+17.1%	Default setting
100	2,000	0.257	+18.4%	Small gain vs. $m = 50$

The burst count has a smaller effect than burst length. Increasing m from 50 to 100 gives only +1.3 percentage points at double the step cost. The marginal gain from additional bursts diminishes because the chain has already explored the best-connected neighborhood of the initial plan.

5.4. Fixed-Budget Comparison

Table 4 compares configurations at exactly 1,000 total steps.

The fixed-budget comparison confirms the key finding: burst length $\ell = 10$ is optimal within a 1,000-step budget. The $(50, 20)$ and $(100, 10)$ configurations perform significantly

Table 4: Fixed-budget comparison ($\ell \cdot m = 1{,}000$ steps) on NC. Baseline: PP = 0.217. The ($\ell = 10, m = 100$) configuration slightly outperforms the default ($\ell = 20, m = 50$).

$(\ell,\ m)$	PP	Improvement	Character
(5, 200)	0.251	+15.7%	Many very short bursts
(10, 100)	0.256	+18.0%	Best fixed-budget setting
(20, 50)	0.254	+17.1%	Default
(50, 20)	0.244	+12.4%	Long bursts, few restarts
(100, 10)	0.235	+8.3%	Approaches 10-seed ConvergenceSweep

worse, confirming that long bursts drift away from good plans.

5.5. Key Finding: Why Short Bursts Win

The sensitivity analysis reveals a clear principle:

> *Short bursts of $\ell = 10$–20 steps outperform long bursts because they stay in the tight neighborhood of good plans. Very long bursts ($\ell = 50+$) drift and resemble single-chain ReCom.*

The intuition: after ℓ steps of ReCom, approximately ℓ/k of the k districts have been touched (each step modifies one pair of adjacent districts). For NC ($k = 14$), $\ell = 20$ modifies ≈ 1.4 districts on average—a genuinely local perturbation. At $\ell = 50$, ≈ 3.6 districts have been modified—the plan has drifted significantly from the starting point. At $\ell = 100$, ≈ 7 of 14 districts have changed, and the plan is no longer recognisably close to the starting point.

The restart mechanism is only valuable when the endpoint is still in the neighborhood of a good plan. For $\ell > 2k$ (where k is the number of districts), the burst covers the entire plan, and the restart provides no concentration benefit. This gives a rule of thumb: $\ell^* \approx k$, i.e., set burst length equal to the number of districts. For NC ($k = 14$), this suggests $\ell^* \approx 14$, consistent with the empirical optimum of $\ell = 10$–20.

5.6. Practical Recommendation

Based on the sensitivity analysis, we recommend:

- Default: $\ell = 20$, $m = 50$ (1,000 total steps, as implemented).
- Alternative for small k ($k \leq 10$): $\ell = k$, $m = 100/\ell \cdot 10$, keeping total steps $\approx 1{,}000$.

- Alternative for large k ($k \geq 30$): $\ell = 30$, $m = 50$ (total 1,500 steps), to ensure each burst covers ≈ 2 districts rather than 1.
- Always use $p = 0.0$ for compactness optimisation. The percentile parameter $p > 0$ is appropriate for ensemble analysis, not for optimization.

6. Conclusion

ShortBurst is a practical search mode for minimum-edge-cut compactness optimisation. By restarting short ReCom chains from each burst's endpoint, it concentrates exploration near good plans without sacrificing the Markov property. The implementation in `redist-cli` as `-search short-burst` achieves 15–22% Polsby-Popper improvement over single-seed METIS, outperforming both multi-seed METIS and ConvergenceSweep under a 1,000-step budget, and approaching BisectionEnsemble performance.

The key correctness requirement is *endpoint retention* rather than within-burst minimization: retaining the burst endpoint preserves the Markov restart property and avoids systematic bias that would make the selected-plan sequence uninterpretable as a well-defined stochastic process.

For the statutory baseline, ShortBurst is most effective combined with ApportionRegions (`-structure prime-factor`): ApportionRegions provides a good initialisation, and ShortBurst refines via concentrated neighbourhood exploration.

Open directions.

1. **Multi-run statistics.** This paper presents single-run results. Phase 2 will provide multi-run statistics (mean, standard deviation, and confidence intervals over 50 independent base seeds) to confirm that the reported improvements are consistent rather than seed-dependent.
2. **Combination with BisectionEnsemble.** The natural next step is to apply ShortBurst as a post-processing step on the output of BisectionEnsemble: BisectionEnsemble optimizes each bisection node, and ShortBurst then refines the final assembled plan.

3. **Burst length theory.** The empirical rule $\ell^* \approx k$ (burst length equal to the number of districts) warrants a formal analysis. The ReCom step modifies $\approx 1/k$ of the plan per step; a burst of k steps covers the plan once on average. Whether this is the optimal coverage rate for neighbourhood concentration is an open question.

4. **Non-uniform stationary distributions.** The stochastic dominance result (Theorem 1) assumes a uniform stationary distribution. For compactness-weighted chains (where plans with higher PP are preferred), the analysis changes: the chain naturally concentrates near good plans, and the Short-Burst advantage may be smaller or larger depending on the weight distribution.

References

Sarah Cannon, Moon Duchin, Dana Randall, and Parker Rule. Voting rights, Markov chains, and optimization. *arXiv*, 2022. arXiv:2207.07690. Introduces Short-Burst optimization for redistricting Markov chains. Short bursts of length ℓ starting from the current plan and restarting from the endpoint concentrate probability near good plans. Original application: partisan outcome analysis. Our use (min-EC compactness) is a natural extension.

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